

# Polyhedron and Spectrahedron on fields with valuations

Corentin Cornou<sup>1</sup>

Simone Naldi<sup>2</sup>

Tristan Vaccon<sup>2</sup>

1. ENS Saclay, Gif-sur-Yvette, France

2. Université de Limoges, CNRS, XLIM, Limoges, France



Université  
de Limoges



**xlim**  
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## 1. General context over $\mathbb{R}$

## 2. Non-archimedean geometry

- ▶ Discrete-valuation fields
- ▶ Objects of interest

## 3. Polyhedron

- ▶ Definition and first examples
- ▶ Stability and applications

## 4. Spectrahedron

- ▶ Positive semi-definite matrices
- ▶ Definitions and first examples
- ▶ Polyannuli

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For  $\ell_1, \dots, \ell_m \in \mathbb{R}[x]_{\leq 1}$  real affine  $n$ -variate functions:

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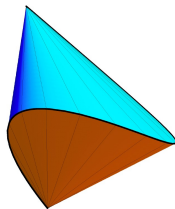
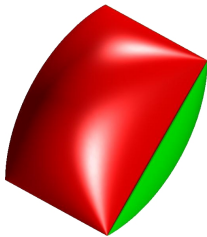
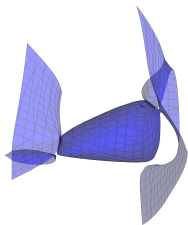
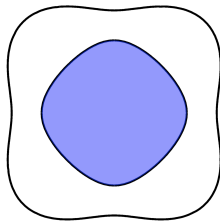
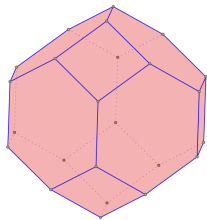
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**Fact:** They are **basic semi-algebraic** sets:  $A(x) \succeq 0$  if and only if all the principal minors of  $A(x)$  are  $\geq 0$ .





# Motivations and properties

Polyhedra  $\rightarrow$  feasible sets of **Linear Programming (LP)**



Spectrahedra  $\rightarrow$  feasible sets of **Semidefinite Programming (SDP)**



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Classical applications from

- ▶ **real algebraic geometry** (positive polynomials vs SOS, 17th Hilbert's problem)
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Several **good properties** besides convex but spectrahedra/SDP are highly more **pathological** than polyhedra/LP:

- ▶ spectrahedra are not closed under linear image
- ▶ solutions only over algebraic extension:  $\{\sqrt{2}\} = \{x \in \mathbb{R} : \begin{bmatrix} x & 1 \\ 2 & x \end{bmatrix} \succeq 0 \text{ and } \begin{bmatrix} 1 & x \\ x & 2 \end{bmatrix} \succeq 0\}$
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## Semidefinite representable sets

The linear projection of a spectrahedron is called a **spectrahedral shadow**:

$$S = \left\{ x \in \mathbb{R}^n : \exists y \in \mathbb{R}^p, A_0 + \sum_{i=1}^n x_i A_i + \sum_{j=1}^p y_j B_j \succeq 0 \right\},$$

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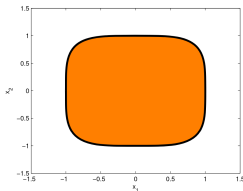


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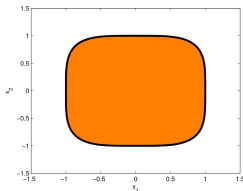
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Not every convex (compact) semialgebraic set is semidefinite representable (Scheiderer, 2018).

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## Discrete-valuation fields

Let  $K$  be a field with:

- ▶ discrete valuation  $v_K : K \rightarrow \mathbb{R} \cup \{+\infty\}$  making it complete,
- ▶ integer ring  $O_K$ ,
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Topology somehow in-between discrete and continuous mathematics.

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## Proposition

- ▶ Any point of  $D_K(\alpha; \lambda)$  is a center of this disk.
- ▶ If two disks have a non-trivial intersection, then one contains the other. As such, any set of disks as a forest / tree-like structure for inclusion.

# Annuli

## Definition (Annulus)

$$\mathcal{C}_K(\alpha, a, b) = \{x \in K \mid a \leq \text{val}(x - \alpha) \leq b\},$$

for  $x \in K$ ,  $a \leq b \in \mathbb{R}$ .

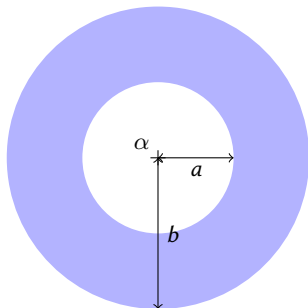


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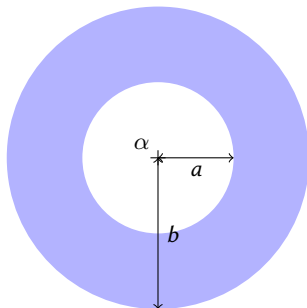


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## Remark

Central object in  $p$ -adic differential equations or in non-archimedean analytic geometry. Used to define series converging on  $\mathcal{C}_K(\alpha, a, b)$ .

## Definition

Let  $\varphi \in K[x]$  be a monic polynomial and let  $\lambda \in \mathbb{Q}$ . Then

$$D_K(\varphi; \lambda) := \{x \in K : v_K(\varphi(x)) \geq \lambda\}$$

is the **discoid** with center  $\varphi$  and radius  $\lambda$  over  $K$ .

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Thanks to recent works with A. Poteaux and M. Weimann, they appear in the clusterization of the roots of polynomials over fields with valuations.

# Visualizing discoids

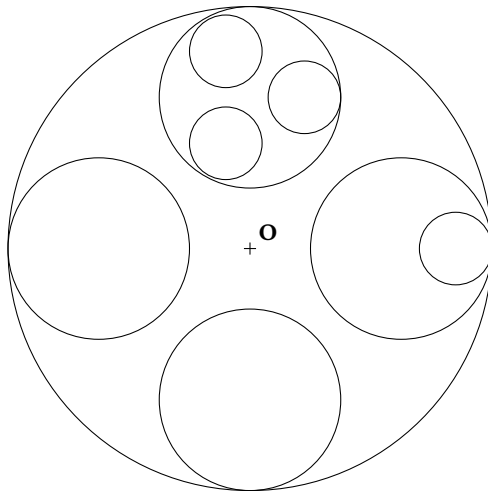
## Proposition

*If  $\varphi$  is irreducible over  $K$ , then  $D_{\overline{K}}(\varphi; \lambda)$  is the union of balls of same radius centered around the roots of  $\varphi$ . The Galois group of  $\overline{K}/K$  acts on those balls.*

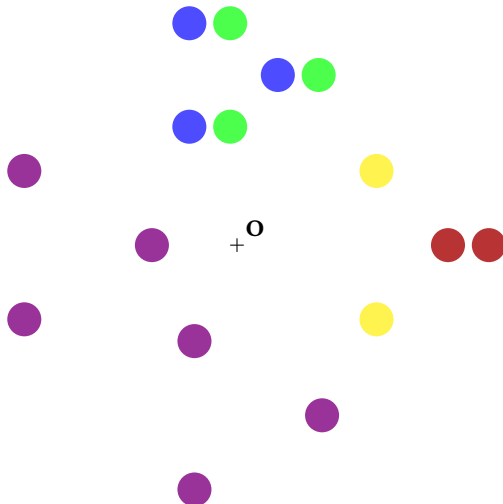
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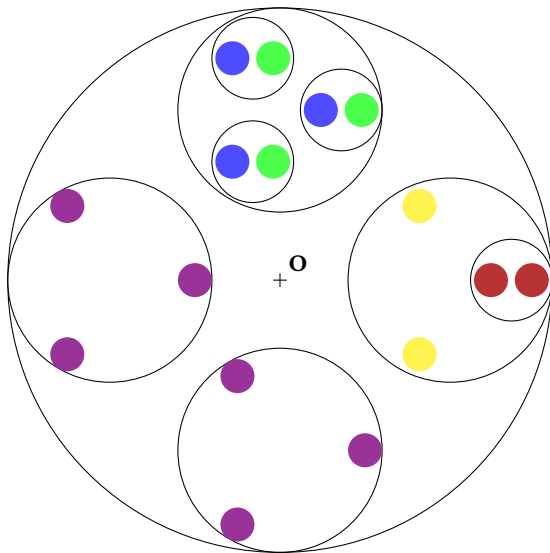


## Visualizing clustering roots in a discrete-valuation field with diskoids

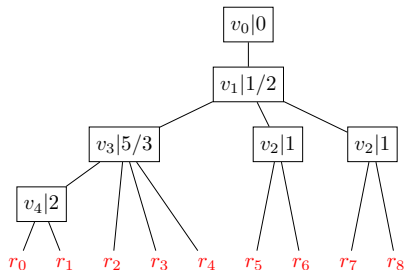




## Visualizing clustering roots in a discrete-valuation field with diskoids



# Clustering roots with diskoids and OM algorithm: over $\mathbb{Q}((t))[x]$



Berkovich skeleton of  $f_1 = (x - t^2)(x - 2t^2)(x^3 - t^5)((x^2 - t)^2 - t^3)$

**In black:** Balls of diskoids defined by:

$v_0$ :  $D(0, 0)$ .

$v_1$ :  $D(\Phi_1, \frac{1}{2})$ , with  $\Phi_1 = x$ .

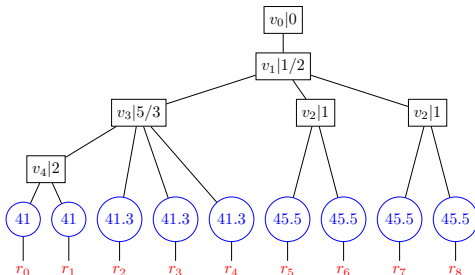
$v_2$ :  $D(\Phi_2, 3/2)$ , with  $\Phi_2 = \Phi_1^2 - t$ .

$v_3$ :  $D(\Phi_3, \frac{5}{3})$ , with  $\Phi_3 = x$ .

$v_4$ :  $D(\Phi_4, 2)$ , with  $\Phi_4 = x$ .

**In red:** Roots.

# Clustering roots of approximation of polynomials in $\mathbb{Q}((t))[x]$



Approximate Berkovich skeleton clustering all the roots of

$$f_1 = (x - t^2)(x - 2t^2)(x^3 - t^5)((x^2 - t)^2 - t^3) + O(t^{50})$$

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$$v_1: D(\Phi_1, \frac{1}{2}), \text{ with } \Phi_1 = x.$$

$$v_2: D(\Phi_2, 3/2), \text{ with } \Phi_2 = \Phi_1^2 - t.$$

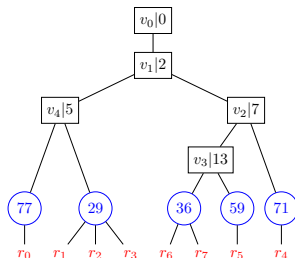
$$v_3: D(\Phi_3, \frac{5}{3}), \text{ with } \Phi_3 = x.$$

$$v_4: D(\Phi_4, 2), \text{ with } \Phi_4 = x.$$

**In blue:** Balls coming from Brink's inequality.

**In red:** Roots.

# Clustering roots of approximation of polynomials in $\mathbb{Q}((t))[x]$ , multiplicity



Approximate Berkovich skeleton clustering all the roots of

$$f_3 = (x - t^2)(x - t^2 - t^5)^3(x + t^2 + t^7)^2(x + t^2 + t^7 + t^{13})(x + t^2) + O(t^{100})$$

**In black:** Balls of given log-radii defined by diskoids defined by the following valuations:

$$v_0: D(0, 0)$$

$$v_1: D(0, 2)$$

$$v_2: D(t^2, 7)$$

$$v_3: D(t^2 + t^7, 13)$$

$$v_4: D(-t^2, 5)$$

**In blue:** Balls coming from Brink's inequality.

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# Non-archimedean polyhedron

## Definition

A *polyhedron* is a subset  $\mathcal{P} \subset K^n$  of the form

$$\mathcal{P} = \left\{ x \in K^n : \begin{array}{ll} \text{val}(\ell_i(x)) \geq 0 & i = 1, \dots, d \\ \text{val}(m_j(x)) = +\infty & j = 1, \dots, e \end{array} \right\},$$

for some  $\ell_1, \dots, \ell_d, m_1, \dots, m_e \in K[x]_1$  (polynomials of degree 1).

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This can be written in matrix form:

$$\mathcal{P} = \{x \in K^n : Ax + v \succeq 0, Bx + w = 0\},$$

for  $A \in K^{d \times n}$ ,  $v \in K^d$ ,  $B \in K^{e \times n}$ , and  $w \in K^e$  such that  $Ax + v = (\ell_1, \dots, \ell_d)^T$  and  $Bx + w = (m_1, \dots, m_e)^T$ .



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## Example

$$D(c, \alpha) = \{x \in K^n : \text{val}(x - c) \geq \alpha\} = \{x \in K^n : \text{val}(\pi^{-\alpha}(x - c))\}.$$

As such, balls are polyhedron, and similarly for polydisks.

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# Projecting polyhedron

## Theorem (direct image (DI))

*Let  $f : K^n \rightarrow K^m$  be an affine map and  $\mathcal{P}$  a polyhedron in  $K^n$ . Then  $f(\mathcal{P})$  is a polyhedron in  $K^m$ .*

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## Idea of proof.

Use the Smith Normal Form of the linear part  $f_0$  of  $f$ .

$$\text{Mat}_B(f_0) = P \begin{pmatrix} \pi^{\alpha_1} & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \pi^{\alpha_2} & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \pi^{\alpha_s} & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 0 \end{pmatrix} Q$$

with  $P, Q \in GL_n(O_K)$ .

Reduce to the case of: translation, isometry, projection.





## Proposition (Characterization of polyhedra)

*A non-empty polyhedron is the image of a polydisc under an affine map.*

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*Polyhedra in  $K$  are either the empty set or balls, possibly with infinite radius.*

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## Corollary

$$O_K^{d \times d} \subset K_+^{d \times d}.$$

## Example

Let

$$M = \begin{bmatrix} \frac{3}{5} + 5 & \frac{4}{5} \\ \frac{4}{5} & -\frac{3}{5} \end{bmatrix} \in \mathbb{Q}_5^{2 \times 2}.$$

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$$\chi_M(T) = T^2 - 5T - 4,$$

Hence

$$M \in (\mathbb{Q}_5)_+^{2 \times 2}$$

$$M \notin (\mathbb{Z}_5)^{2 \times 2}.$$

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$$\mathcal{S} = \mathcal{L} \cap K_+^{d \times d} = \left\{ X \in K^{d \times d} : \exists x \in K^n, X = A_0 + \sum_{i=1}^n x_i A_i, X \succeq 0 \right\}$$

where  $\mathcal{L} = A_0 + \text{span}(A_1, \dots, A_n) \subset K^{d \times d}$  is an affine space, for some  $A_0, A_1, \dots, A_n \in K^{d \times d}$ .

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A spectrahedron in  $K^n$  is any affine section of a set of the form

$$\mathcal{S}_A = \varphi_A^{-1}(K_+^{d \times d}) = \{x \in K^n : A_0 + x_1 A_1 + \dots + x_n A_n \succeq 0\}$$

for some  $d \in \mathbb{N}$ , where  $\varphi_A : K^n \rightarrow \mathcal{L} \subset K^{d \times d}$  is the map  $\varphi_A(x) = A_0 + x_1 A_1 + \dots + x_n A_n$ , and  $\mathcal{L} = A_0 + \text{span}(A_1, \dots, A_n)$ .

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Assume  $\mathcal{P} = L \cap M$  be a polyhedron in  $K^n$ , with  $L = \{x \in K^n : \text{val}\ell_i(x) \geq 0, i = 1, \dots, d\}$  and  $M = \{x \in K^n : \text{val}m_j(x) = +\infty, j = 1, \dots, e\}$ , for some  $d, e \in \mathbb{N}$ .



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Let  $A_0, A_1, \dots, A_n \in K^{d \times d}$  be such that

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One has  $L = \mathcal{S}_A = \varphi_A^{-1}(K_+^{d \times d})$ . Since  $M \subset K^n$  is affine, then  $\mathcal{P}$  is a spectrahedron in  $K^n$ . □

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The following spectrahedron is not a polyhedron:

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which is not a ball in  $K$ . Therefore,  $\mathcal{S}$  is not a polyhedron.  $\square$

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for some integer  $d$ , matrices  $A_0$  and  $A_1$  in  $K^{d \times d}$ , and  $a_i$  the coefficient of the corresponding characteristic polynomial. The intersection of two diskoids is either empty or a diskoid. Consequently, spectrahedra of  $K$  are diskoids. □



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## Definition (Semidefinite-representable set)

A set  $\mathcal{R} \subset K^n$  is called *semidefinite-representable (SDR)* if there exist  $m, d \in \mathbb{N}$  and a spectrahedron  $\mathcal{S} \subset K^{n+m}$  such that

$$\mathcal{R} = \text{proj}_{\leq n}(\mathcal{S})$$

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If  $m \geq 1$ , we can also write that  $\mathcal{R}$  is a spectrahedral shadow.

# Annuli and shadows

## Theorem

*Annuli in  $K$  such as  $\mathcal{C}_K(0, a, b) = \{x \in K \mid a \leq \text{val}(x) \leq b\}$  are spectrahedral shadows.*

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$$M(x, y) := \begin{pmatrix} \begin{bmatrix} \pi^{-a}x & 0 & 0 & 0 \\ 0 & \pi^b y & 0 & 0 \\ 0 & 0 & \pi^{-1} & -\pi^{-1}x \\ 0 & 0 & \pi^{-1}y & -\pi^{-1} \end{bmatrix} \\ \succeq 0. \end{pmatrix}$$

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Thus  $\text{val}(x) + \text{val}(y) = \text{val}(xy) = \text{val}(-1) = 0$ . Therefore  $\text{val}(x) = -\text{val}(y) \leq -(-b) = b$  and  $x \in \mathcal{C}_K(0, a, b)$ .



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Annuli in  $K$  such as  $\mathcal{C}_K(0, a, b) = \{x \in K \mid a \leq \text{val}(x) \leq b\}$  are spectrahedral shadows.

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$$M(x, y) := \begin{pmatrix} \begin{bmatrix} \pi^{-a}x & 0 & 0 & 0 \\ 0 & \pi^b y & 0 & 0 \\ 0 & 0 & \pi^{-1} & -\pi^{-1}x \\ 0 & 0 & \pi^{-1}y & -\pi^{-1} \end{bmatrix} \end{pmatrix} \succeq 0.$$

Then  $M \succeq 0$  if and only if  $(x, y)$  is a solution of the system of inequalities:

$$\begin{cases} \text{val}(\pi^{-a}x) \geq 0 \\ \text{val}(\pi^b y) \geq 0 \\ \text{val}(\pi^{-1} - \pi^{-1}) = +\infty \geq 0 \\ \text{val}(\pi^{-2}(xy - 1)) \geq 0 \end{cases} \iff \begin{cases} \text{val}(x) \geq a \\ \text{val}(y) \geq -b \\ \text{val}(xy - 1) \geq 2 \end{cases}$$

Thus  $\text{val}(x) + \text{val}(y) = \text{val}(xy) = \text{val}(-1) = 0$ . Therefore  $\text{val}(x) = -\text{val}(y) \leq -(-b) = b$  and  $x \in \mathcal{C}_K(0, a, b)$ . The converse is clear and  $\mathcal{C}_K(0, a, b)$  is the proj. of the spectrahedra defined by  $M$ .

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# Annuli and spectrahedron

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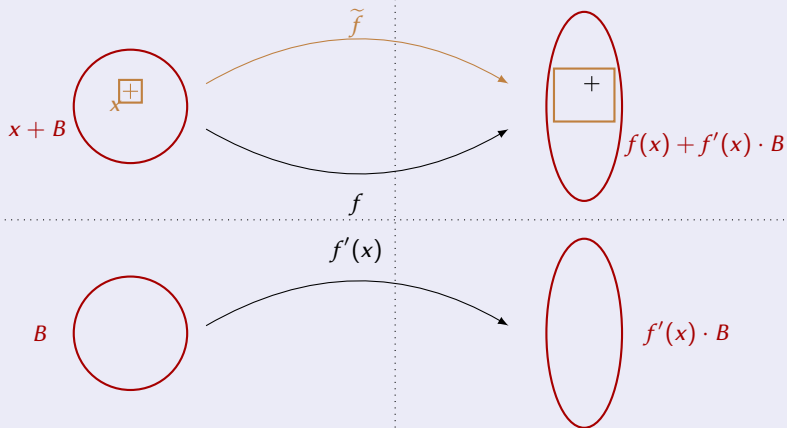
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Thank you for your attention

Thank you

$$x + O(p^{N'})$$

$$y + O(p^{M'}) \subset f(x) + O(p^N)$$



## OM algorithms and diskoids

- ▶ **PVW2025**, *On OM algorithms and Cluster Pictures*, A. Poteaux, T. Vaccon, M. Weimann, 2025.
- ▶ **PVW2026**, *On OM algorithms and Cluster Pictures II: handling small precision*, A. Poteaux, T. Vaccon, M. Weimann, 2026.

## Polyhedron and spectrahedron over valued fields

- ▶ **CNV2025**: *On semidefinite-representable sets over valued fields*, C. Cornou, S. Naldi, T. Vaccon, 2026

## A tropical approach to spectrahedra in a non-archimedean context (unrelated)

- ▶ **AGS2020**: *Tropical Spectrahedra*, X. Allamigeon, S. Gaubert, M. Skomra, 2020